Curriculum Vitae of **Ishtiaque Ahmed Navid**

Device Engineer
Advanced Logic Technology Platform
IBM Research, Albany, NY

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RESEARCH INTERESTS:

Semiconductor Logic Device | Artificial Photosynthesis | Optoelectronics| III-Nitride Compound Semiconductors | Low-Dimensional Materials | Thermal and Electronic Transport

EDUCATION:

- Doctor of Philosophy in Electrical and Computer Engineering (ECE), University of Michigan, Ann Arbor, December 2024, **GPA:4.00/4.00**
- Master of Science in Electrical and Electronic Engineering (EEE), Bangladesh University of Engineering & Technology (BUET), November 2018, **GPA:3.92/4.00, 2nd highest** in the department of EEE, BUET
- Bachelor of Science in Electrical and Electronic Engineering (EEE), Bangladesh University of Engineering & Technology (BUET), March 2016, **GPA:3.92/4.00 (among top 5%)** in the department of EEE, BUET

INDUSTRY WORK EXPERIENCE:

Device Engineer (August 2024-Present)
IBM Research, Albany, NY

Projects:

- 2 nm Gate-All-Around (GAA) Nanosheet Field Effect Transistor (FET) Technology Research and Development
- Multiple-Vt Solutions in Nanosheet Technology for High Performance and Low Power Applications
- eFuse Development for Advanced Technology Node in Complementary Metal-Oxide-Semiconductor (CMOS)

Responsibilities:

- Design semiconductor devices with a focus on performance optimization and innovation.
- Work with the simulation modeling team to understand device behavior and predict performance.
- Collaborate with fabrication teams to develop and refine processes for fabricating advanced semiconductor devices.
- Work with the test team to perform electrical and physical characterization of semiconductor devices.
- Use statistical and computational tools to interpret results and guide further research.
- Prepare reports and presentations for internal stakeholders, conferences, and publications.

RESEARCH EXPERIENCE:

University of Michigan, Ann Arbor **September 2019-September 2024**
Graduate Student (Ph.D.) Research Assistant | advised by **Professor Zetian Mi**

- **High-efficiency and long-term stability of solar water splitting in InGaN nanowire photocatalyst:** Design and optimization of epitaxial growth, reaction conditions, and protection architecture in p-type GaN/InGaN heterostructure nanowires for achieving record-high efficiency and long-term stability in photocatalytic (PC) solar water splitting
- **Deep-nano micro-network structures for optoelectronics and photocatalysis:** Molecular beam epitaxy (MBE), optical characterization, and photocatalytic property investigation of GaN/InGaN micro-network deep-nano structure
- **Design and performance optimization of InGaN/Si solar cell-based tandem photocathode:** Fabrication, MBE growth, characterization, and device testing of p-type InGaN nanowire/Si solar cell-based double junction tandem photocathode for photoelectrochemical (PEC) solar water splitting
- **Dilute anion-doped III-nitride nanostructures for solar fuel applications:** Design, material epitaxy, and property investigation of dilute Bi-doped GaN-based nanostructures

- **Single junction GaN photocathode optimization and co-catalyst engineering for PEC reactions:** Fabrication, MBE growth, co-catalyst engineering, and characterization of n-type GaN single junction photocathode for PEC solar water splitting and CO_2 reduction
- **Polarization and strain engineering in InGaN-based micro-LEDs:** Selective area growth (SAG), strain engineering, and characterization of N-polar GaN/InGaN/AlGaN nanowire green micro-LEDs on Si substrate

Bangladesh University of Engineering and Technology (BUET) **February 2015-August 2019**
Graduate (M.Sc.) and Undergraduate (B.Sc.) Researcher | advised by **Professor Samia Subrina**

- **Thermal transport modeling in novel 2D nanomaterials:** Pristine and defect-induced graphene nanoribbons, pristine and strain-engineered hexagonal boron nitride nanoribbons | Molecular Dynamics Simulation
- **Electronic property simulation of novel 2D nanomaterials:** Pristine and C/Si-doped stanene nanostructures | Density Functional Theory (DFT) Simulation

ACADEMIC/TEACHING EXPERIENCE:

- **Graduate Student Grader** (January 2024-April 2024)
  University of Michigan, Ann Arbor (EECS)
  EECS 620 (Electronic and Optical Properties of Semiconductors)
  Primary Instructor: **Professor Zetian Mi**
- **Graduate Student Instructor** (August 2022-December 2022)
  University of Michigan, Ann Arbor (EECS)
  EECS 320 (Introduction to Semiconductor Devices)
  Primary Instructor: **Professor Stephen R. Forrest**
- **Lecturer** (October 2016-August 2019)
  United International University, Bangladesh (EEE)
  Courses Instructed: Electrical Properties of Materials, Electrical Circuit-I**,** Microprocessor & Interfacing Laboratory, Electronics Laboratory, Electrical Circuit-I Laboratory

PUBLICATIONS: (>2300 citations, i10-index: **25,** H-index: **19** in **Google Scholar)**

- **Journal Articles:**

[32] Bingxing Zhang, Zhiheng Zhao, Yuyang Pan, Yishan Shen, Zhengwei Ye, Li Wang, **Ishtiaque Ahmed Navid**, Kai Sun, Frank E Osterloh, Theodore Norris, and Zetian Mi "Photocatalytic Overall Water Splitting at the Integrated Rh-MoRhO_x Cluster Heterostructure on InGaN/GaN Nanowires", **Angewandte Chemie**, 2026 (Vol. 138, Issue 2), Publisher: Wiley, DOI: 10.1002/ange.202520831

[31] Ishtiaque Ahmed Navid, Wan Jae Dong, and Zetian Mi "Metal/Semiconductor Materials for Photothermal Catalysis", **Artificial Photosynthesis**, 2026 (Vol. 2, Issue 1), Publisher: American Chemical Society (ACS), DOI: 10.1021/aps.5c00001

[30] Ishtiaque Ahmed Navid, Zhengwei Ye, Yuyang Pan, Yifan Shen, Theodore Norris, and Zetian Mi "Long-Term Stability of GaN-based Photocatalyst Nanostructures through Dynamic Oxide Protection", **Journal of Materials Chemistry A**, 2025 (Vol. 13, Issue 37), Publisher: Royal Society of Chemistry (RSC), DOI: 10.1039/d5ta02771c

[29] W. J. Dong, J. P. Menzel, Z. Ye, Z. Long, **I. A. Navid**, K. R. Yang, Y. Xiao, V. S. Batista, and Z. Mi, "Nitrate Reduction to Ammonia Catalyzed by GaN/Si Photoelectrodes with Metal Clusters", **Nature Communications**, 2025 (Vol. 16, Issue 1), Publisher: Nature Portfolio, DOI: 10.1038/s41467-025-58372-7

[28] Wan Jae Dong, Jan Paul Menzel, Zhengwei Ye, Zhuoran Long, **Ishtiaque Ahmed Navid**, Victor S Batista, and Zetian Mi "Synergistic Metal-Support Interactions in Au/GaN Catalysts for Photoelectrochemical Nitrate Reduction to Ammonia", **Small,** 2025 (Vol. 21, Issue 11), Publisher: Wiley, DOI: 10.1002/smll.202412089

[27] Ishtiaque Ahmed Navid, Yifan Shen, Peng Zhou, Jonathan Schwartz, Yin Min Goh, Tao Ma, Robert Hovden, Theodore Norris, and Zetian Mi "Molecular beam epitaxy, photocatalytic solar water splitting, and carrier dynamics of InGaN micro-network deep-nano structures", **AIP Advances,** 2025 (Vol. 15, Issue 1), Publisher: American Institute of Physics (AIP), DOI: 10.1063/5.0212409

[26] Zhengwei Ye, Zhuoran Long, Bingxing Zhang, **Ishtiaque Ahmed Navid**, Paul Menzel, Yifan Shen, Shubham Mondal, Facheng Guo, Theodore B Norris, Victor S Batista, Zetian Mi "Photocatalytic Conversion of Methane to

Ethane and Propane Using Cobalt-Cluster-Activated GaN Nanowires", **Angewandte Chemie,** 2025 (Vol. 137, Issue 14), Publisher: Wiley, DOI: 10.1002/ange.202500158

[25] Yujie Liu, **Ishtiaque Ahmed Navid**, Zetian Mi, and Emmanouil Kioupakis "Selective incorporation of antimony into gallium nitride", **Applied Physics Letters,** 2024 (Vol. 125, Issue 12), Publisher: American Institute of Physics (AIP), DOI: 10.1063/5.0219021

[24] Bingxing Zhang, Peng Zhou, Zhengwei Ye, **Ishtiaque Ahmed Navid**, Yuyang Pan, Yixin Xiao, Kai Sun, and Zetian Mi "Interfacially coupled Cu-cluster/GaN photocathode for efficient CO_2 to ethylene conversion", **Nature Synthesis,** 2024, Publisher: Nature Portfolio, DOI: 10.1038/s44160-024-00648-9

[23] Zhengwei Ye, Ke Yang, Bingxing Zhang, **Ishtiaque Ahmed Navid**, Yifan Shen, Yixin Xiao, Alexandre Pofelski, Gianluigi A Botton, Tao Ma, Shubham Mondal, Theodore Norris, Victor S. Batista, and Zetian Mi "A synergetic cocatalyst for conversion of carbon dioxide, sunlight, and water into methanol", **Proceedings of the National Academy of Sciences**, 2024 (Vol. 121, Issue 35), Publisher: National Academy of Sciences, DOI: 10.1073/pnas.2408183121

[22] Wan Jae Dong, Zhengwei Ye, Songtao Tang, **Ishtiaque Ahmed Navid**, Yixin Xiao, Peng Zhou, Bingxing Zhang, Yuyang Pan, and Zetian Mi "Concentrated Solar Light Photoelectrochemical Water Splitting for Low Cost, High Yield, and Stable Hydrogen Production", **Advanced Science,** 2024 (Vol. 11, Issue 26), Publisher: Wiley, DOI: 10.1002/advs.202309548

[21] W. J. Dong, J. P. Menzel, Z. Ye, **I. A. Navid**, P. Zhou, K. R. Yang, V. S. Batista, and Z. Mi, "Photoelectrochemical Urea Synthesis from Nitrate and Carbon Dioxide on GaN Nanowires", **ACS Catalysis**, 2024 (Vol. 14, Issue 4), Publisher: American Chemical Society (ACS), DOI: 10.1021/acscatal.3c04264

[20] Ishtiaque Ahmed Navid, Yujie Liu, Yuyang Pan, Kai Sun, Emmanouil Kioupakis, and Zetian Mi "Structural and Optical Characterization of Dilute Bi-doped GaN Nanostructures Grown by Molecular Beam Epitaxy", **APL Materials**, 2024 (Vol. 12, Issue 2), Publisher: American Institute of Physics (AIP), DOI: 10.1063/5.0180567

[19] Peng Zhou, Zhengwei Ye, Ke Yang, **Ishtiaque Ahmed Navid**, Alexandre Pofelski, Gianluigi A Botton, Tao Ma, Yixin Xiao, Shubham Mondal, Victor S. Batista, and Zetian Mi "Water-promoted selective photocatalytic methane oxidation for methanol production", **Chemical Science**, 2024, (Vol. 15), Publisher: Royal Society of Chemistry (RSC), DOI: 10.1039/D3SC02567E

[18] Baowen Zhou, Yongjin Ma, Pengfei Ou, Zhengwei Ye, Xiao-Yan Li, Srinivas Vanka, Tao Ma, Haiding Sun, Ping Wang, Peng Zhou, Jason K Cooper, Yixin Xiao, **Ishtiaque Ahmed Navid**, Jun Pan, Jun Song, and Zetian Mi, "Light-driven synthesis of C_2H_6 from CO_2 and H_2O on a bimetallic AuIr composite supported on InGaN nanowires", **Nature Catalysis**, 2023, (Vol. 6, Issue 11), Publisher: Nature Portfolio, DOI: 10.1038/s41929-023-01023-1

[17] Yixin Xiao, Xianghua Kong, Srinivas Vanka, Wanjae Dong, Guosong Zeng, Zhengwei Ye, Kai Sun, **Ishtiaque Ahmed Navid**, Baowen Zhou, Francesca M. Toma, Hong Guo, and Zetian Mi, "Oxynitrides enabled photoelectrochemical water splitting with over 3,000 hrs stable operation in practical two-electrode configuration", **Nature Communications**, 2023, (Vol. 14, Article 2047), Publisher: Nature Portfolio, DOI: 10.1038/s41467-023-37754-9

[16] Wan Jae Dong, Yixin Xiao, Ke R. Yang, Zhengwei Ye, Peng Zhou, **Ishtiaque Ahmed Navid**, Victor S. Batista, and Zetian Mi, "Pt Nanoclusters on GaN Nanowires for Solar-assisted Seawater Hydrogen Evolution", **Nature Communications**, 2023, (Vol. 14, Article 179), Publisher: Nature Portfolio, DOI: 10.1038/s41467-023-35782-z

[15] Peng Zhou, **Ishtiaque Ahmed Navid**, Yongjin Ma, Yixin Xiao, Ping Wang, Zhengwei Ye, Baowen Zhou, and Zetian Mi "Solar-to-hydrogen efficiency of more than 9% in unassisted photocatalytic overall water splitting", **Nature**, 2023 (Vol. 613, Page 66-70), Publisher: Nature Portfolio, DOI: 10.1038/s41586-022-05399-1

[14] Yuanpeng Wu, Yixin Xiao, **Ishtiaque Navid**, Kai Sun, Yakshita Malhotra, Ping Wang, Ding Wang, Yuanxiang Xu, Ayush Pandey, Reddeppa Maddakka, Walter Shin, Jiangnan Liu, Jungwook Min, and Zetian Mi, "InGaN Micro-light-emitting Diodes Monolithically Grown on Si: Achieving Ultrastable Operation through Polarization and Strain Engineering", **Light: Science and Applications**, 2022 (Vol. 11, Article 294), Publisher: Nature Portfolio, DOI: 10.1038/s41377-022-00985-4

[13] Peng Zhou, **Ishtiaque Ahmed Navid**, Yixin Xiao, Zhengwei Ye, Wanjae Dong, Ping Wang, Kai Sun, and Zetian Mi "Metal-support Interaction-promoted Photothermal Catalytic Methane Reforming into Liquid Fuels", **The Journal of Physical Chemical Letters**, 2022 (Vol. 13, Issue 34), Publisher: American Chemical Society (ACS), DOI: 10.1021/acs.jpclett.2c01459

[12] Wan Jae Dong*, **Ishtiaque Ahmed Navid***, Yixin Xiao, Tae Hyung Lee, Jin Wook Lim, Donghwa Lee, Ho Won Jang, Jong-Lam Lee, and Zetian Mi "Bi Catalysts Supported on GaN Nanowires towards Efficient Photoelectrochemical CO_2 Reduction Using Si Photocathodes", **Journal of Materials Chemistry A**, 2022 (Vol. 10, Issue 14), Publisher: Royal Society of Chemistry (RSC), DOI: 10.1039/D2TA00032F

[11] Yixin Xiao, Srinivas Vanka, Tuan Anh Pham, Wan Jae Dong, Yi Sun, Xianhe Liu, **Ishtiaque Ahmed Navid**, Hamed Hajibabaei, Thomas Hamann, Tadashi Ogitsu, and Zetian Mi "Crystallographic Effects of GaN Nanostructures in Photoelectrochemical Reaction", **Nano Letters**, 2022 (Vol. 22, Issue 6), Publisher: American Chemical Society (ACS), DOI: 10.1021/acs.nanolett.1c04220

[10] Wan Jae Dong, Peng Zhou, Yixin Xiao, **Ishtiaque Ahmed Navid**, Jong-Lam Lee, and Zetian Mi "Silver Halide Catalysts on GaN Nanowires/Si Heterojunction Photocathodes for CO_2 Reduction to Syngas at High Current Density", **ACS Catalysis,** 2022 (Vol. 12, Issue 4), Publisher: American Chemical Society (ACS), DOI:10.1021/acscatal.1c04904

[9] Ishtiaque Ahmed Navid, Ayush Pandey, Yin Min Goh, Jonathan Schwartz, Robert Hovden, and Zetian Mi "GaN-Based Deep-nano Structures: Break the Efficiency Bottleneck of Conventional Nanoscale Optoelectronics", **Advanced Optical Materials**, 2022 (Vol. 10, Issue 5), Publisher: Wiley, DOI: 10.1002/adom.202102263

[8] Wan Jae Dong*, **Ishtiaque Ahmed Navid***, Yixin Xiao*, Jin Wook Lim, Jong-Lam Lee, and Zetian Mi "CuS-coated GaN Nanowires on Silicon Photocathodes for Converting CO_2 Mixture Gas to HCOOH", **Journal of the American Chemical Society,** 2021 (Vol. 143, Issue 27), Publisher: American Chemical Society (ACS), DOI: 10.1021/jacs.1c02139

[7] Ishtiaque Ahmed Navid*, Srinivas Vanka*, Rasha A. Awni, Yixin Xiao, Zhaoning Song, Yanfa Yan, and Zetian Mi "On the Design and Performance of InGaN/Si Double-Junction Photocathodes", **Applied Physics Letters**, 2021 (Vol. 118, Issue 24), Publisher: American Institute of Physics (AIP), DOI: 10.1063/5.0050708

[6] Srinivas Vanka, Baowen Zhou, Rasha A Awni, Zhaoning Song, Faqrul A Chowdhury, Xuedong Liu, Hamed Hajibabaei, Wen Shi, Yixin Xiao, **Ishtiaque A Navid**, Ayush Pandey, Rong Chen, Gianluigi A Botton, Thomas W Hamann, Dunwei Wang, Yanfa Yan, and Zetian Mi "InGaN/Si Double-Junction Photocathode for Unassisted Solar Water Splitting", **ACS Energy Letters**, 2020 (Vol. 5, Issue 12), Publisher: American Chemical Society (ACS), DOI: 10.1021/acsenergylett.0c01583

[5] Ishtiaque Ahmed Navid and Samia Subrina "Thermal transport characterization of carbon and silicon doped stanene nanoribbon: an equilibrium molecular dynamics study"; **RSC Advances,** 2018 (Vol. 8, Issue 55), Publisher: Royal Society of Chemistry (RSC), DOI: 10.1039/C8RA06156D

[4] Ishtiaque Ahmed Navid, Asir Intisar Khan, and Samia Subrina "Impact of tensile strain on thermal conductivity of zigzag hexagonal boron nitride nanoribbon: An equilibrium molecular dynamics simulation", **Materials Research Express,** 2018 (Vol. 5, Issue 2), Publisher: Institute of Physics (IOP), DOI: 10.1088/2053-1591/aaaa89

[3] Asir Intisar Khan, **Ishtiaque Ahmed Navid**, Maliha Noshin, and Samia Subrina "Thermal Transport Characterization of Hexagonal Boron Nitride Nanoribbons Using Molecular Dynamics Simulation", **AIP Advances,** 2017 (Vol. 7, Issue 10), Publisher: American Institute of Physics (AIP), DOI: 10.1063/1.4997036

[2] Maliha Noshin, Asir Intisar Khan, **Ishtiaque Ahmed Navid**, H. M. Ahsan Uddin, and Samia Subrina "Impact of vacancies on the thermal conductivity of graphene nanoribbons: A molecular dynamics simulation study", **AIP Advances,** 2017 (Vol. 7, Issue 1), Publisher: American Institute of Physics (AIP), DOI: 10.1063/1.4974996

[1] Asir Intisar Khan*, **Ishtiaque Ahmed Navid***, Maliha Noshin, H. M. Ahsan Uddin, Fahim Ferdous Hossain, and Samia Subrina "Equilibrium Molecular Dynamics (MD) Simulation Study of Thermal Conductivity of Graphene Nanoribbon: A Comparative Study on MD Potentials", **Electronics:** Special issue Two-Dimensional Electronics-Prospects and Challenges, 2015 (Vol. 4, Issue 4), Publisher: Multidisciplinary Digital Publishing Institute (MDPI), DOI: 10.3390/electronics4041109

*__Contributed equally__

- **Articles/Presentations in Conference Proceedings (Refereed):**

[14] Josey Hanish, Yifan Shen**,** Yuyang Pan**, Ishtiaque A. Navid**, Zetian Mi, and Theodore Norris "Dynamics of Charge Separation in Photocatalytic InGaN Nanostructures", CLEO: Applications and Technology, 2025, Long Beach, California, USA

[13] Yifan Shen, Josey Hanish, **Ishtiaque A. Navid**, Yuyang Pan, Zetian Mi, and Theodore Norris "Ultrafast Charge Carrier Separation in InGaN Photocatalytic Nanostructures", CLEO: Fundamental Science, 2024, Charlotte, North Carolina, USA

[12] Yujie Liu, **Ishtiaque A. Navid**, Zetian Mi, and Emmanouil Kioupakis "Controllable incorporation of dopant and dilute alloy elements in nitrides", International Conference on Defects in Semiconductors 2023, Rehoboth Beach, Delaware, USA

[11] Ishtiaque Ahmed Navid, Yujie Liu, Emmanouil Kioupakis, and Zetian Mi "Structural and Optical Characterization of Dilute Bi-doped GaN Nanowires Grown by Molecular Beam Epitaxy", APS March Meeting 2023, Las Vegas, Nevada, USA

[10] Yujie Liu, **Ishtiaque Ahmed Navid**, Zetian Mi, and Emmanouil Kioupakis "Theoretical Interpretations of Sb Incorporation in GaN", APS March Meeting 2023, Las Vegas, Nevada, USA

[9] Ishtiaque Ahmed Navid, Ayush Pandey, Yin Min Goh, Jonathan Schwartz, Robert Hovden, and Zetian Mi "GaN-Based Deep-nano Structures: Break the Efficiency Bottleneck of Conventional Nanoscale Optoelectronics", Compound Semiconductor Week (CSW) 2022, Ann Arbor, Michigan, USA

[8] Yixin Xiao, Srinivas Vanka, Wanjae Dong, Yi Sun, Xianhe Liu, **Ishtiaque Ahmed Navid**, Hamed Hajibabaei, Thomas W. Hamann, and Zetian Mi "Polarity Effects of GaN Nanostructures in Photoelectrochemical Reaction", Compound Semiconductor Week (CSW) 2022, Ann Arbor, Michigan, USA

[7] Ishtiaque Ahmed Navid, Yin Min Goh, Jonathan Schwartz, Ayush Pandey, Peng Zhou, Robert Hovden, and Zetian Mi "Molecular Beam Epitaxy, Carrier Dynamics, and Photocatalytic Properties of III-nitride Nanowall Network", 63rd Electronic Materials Conference (63rd EMC) 2021

[6] Yixin Xiao, Srinivas Vanka, **Ishtiaque Ahmed Navid**, and Zetian Mi "Non-Polar Surfaces of GaN Nanostructures in Photoelectrochemical Water Splitting", 63rd Electronic Materials Conference (63rd EMC) 2021

[5] Ishtiaque Ahmed Navid, Md Asaduz Zaman Mamun, Mehedi Hasan, and Samia Subrina " Thermal Transport in Carbon Doped Armchair Stanene Nanoribbons: A Molecular Dynamics Study", IEEE Region 10 Symposium (TENSYMP) 2020

[4] Julker Neyen Sampad, Faiaz Khan, **Ishtiaque Ahmed Navid,** and Samia Subrina "Transport Characteristics of TMDC materials for Triple Material Double Gate MOSFET", IEEE WIECON-ECE 2018 – IEEE International WIE Conference on Electrical and Computer Engineering, Pattaya, Thailand

[3] Nusrat Jahan, **Ishtiaque Ahmed Navid,** and Samia Subrina "Thermal Conductivity of Silicene Nanoribbons: An Equilibrium Molecular Dynamics Study", IEEE WIECON-ECE 2018 – IEEE International WIE Conference on Electrical and Computer Engineering, Pattaya, Thailand

[2] Maliha Noshin, Asir Intisar Khan, **Ishtiaque Ahmed Navid,** and Samia Subrina "Thermal Transport in Defected Armchair Graphene Nanoribbon: A Molecular Dynamics Study", TENCON 2017 – 2017 IEEE Region 10 Conference, Penang, Malaysia

[1] Asir Intisar Khan, **Ishtiaque Ahmed Navid**, Fahim Ferdous Hossain, Maliha Noshin, and Samia Subrina "A Molecular Dynamics Study on Thermal Conductivity of Armchair Graphene Nanoribbon", TENCON 2016 – 2016 IEEE Region 10 Conference, Singapore

PATENTS:

[4] Ishtiaque Navid, Zetian Mi, "CAPPING LAYER-BASED PHOTOCATALYTIC DEVICE PROTECTION", US Patent App. 63/69, 227 (in progress)

[3] Ishtiaque Navid, Ayush Pandey, Zetian Mi, "MICRO-NETWORK INTERCONNECTED NANOSTRUCTURES", US Patent App. 18/703,499 (2025)

[2] Yakshita Malhotra, Yifan Shen, Yuanpeng Wu, Yifu Guo, Yixin Xiao, Kai Sun, **Ishtiaque Navid**, Theodore Norris, Zetian Mi, "LEDS and METHODS OF MANUFACTURE", US Patent App. 18/648, 202 (2024)

[1] Yuanpeng Wu, Yixin Xiao, **Ishtiaque Navid**, Yakshita Malhotra, Zetian Mi, "Light emitting device", US Patent App. 18/200, 375 (2023)

AWARDS AND SCHOLARSHIPS:

- **Richard F. and Eleanor A. Towner Prize for Distinguished Academic Achievement,** from the College of Engineering at the University of Michigan, recognizing outstanding graduate students who excel in research, leadership, and academic performance (2023)
- **Rackham Predoctoral Fellowship,** awarded as an outstanding doctoral candidate working on a dissertation that is unusually creative, ambitious, and impactful (2023)
- **SPIE Optics and Photonics Education Scholarship,** awarded for potential contributions in the fields of optics and photonics (2023)

- **Rackham Conference Travel Grant** for APS March Meeting (2023)
- **John B. Fenn, Sr. Foundation Endowed Scholarship**, one of the ten worldwide winners for the contribution to the field of vacuum coating technology, issued by the Society of Vacuum Coaters Foundation (2023)
- **APS Topical Group on Energy Research and Applications (GERA) Travel Grant** for APS GERA energy research workshop (2023)
- **Best Paper Award** for the paper titled "Thermal Conductivity of Silicene Nanoribbons: An Equilibrium Molecular Dynamics Study", IEEE International WIE Conference on Electrical and Computer Engineering, Pattaya, Thailand (2018)
- **Runner-up Award,** EEE Day Project Show 2015, organized by IEEE Bangladesh Section, Bangladesh University of Engineering and Technology, Dhaka, Bangladesh (2015)
- **Dean's List Scholarship** from Bangladesh University of Engineering and Technology (For all undergraduate academic years, 2012-2015)
- **University Merit Scholarship**, Bangladesh University of Engineering and Technology (For excellent academic performance in seven out of eight undergraduate academic semesters, 2012-2015)
- **Admission Test Excellence Award** for obtaining a merit position 24th among ~10000 participants (top ~0.2%) in the nationwide admission test/entrance exam of Bangladesh University of Engineering and Technology (2011)
- **3rd Prize, Physics Olympiad**, Education and Cultural Conference-2009, Dhaka, Bangladesh (2009)
- **2nd Prize, Science Project Show (Senior Group)**, 30th National Science, Information and Technology Week, Dhaka, Bangladesh (2009)
- **Perfect Attendance Award** from Notre Dame College, Dhaka, Bangladesh for 100 percent attendance in higher secondary classes (2008-2009)
- **Half-free studentship** for excellent academic performance in higher secondary classes, Notre Dame College, Dhaka, Bangladesh (2009)
- **National Scholarship** in Secondary School Certificate (SSC) examination from Bangladesh Education Board, Dhaka, Bangladesh (2008)
- **National Scholarship** in Junior Scholarship Examination (8th Grade) from Bangladesh Education Board, Dhaka, Bangladesh (2005)

LEADERSHIP AND VOLUNTEERING EXPERIENCE:

- **Mentor, Michigan Semiconductor Hands-On Research Experience (M-SHORE) program (May 2023-August 2023)**
  Worked as a mentor for undergraduate students in the M-SHORE program funded by the Research Experience for Undergraduates (REU) program of the National Science Foundation (NSF)
- **Industry Chair, The Optics Society at the University of Michigan (September 2022-April 2023)**
  Successfully organized an industry seminar focusing on job and internship opportunities in the semiconductor industry (December 2022).
- **Volunteer, Xplore Engineering Workshop organized by The Optics Society at the University of Michigan (July 2022)**
  Walked through a premade presentation and assisted elementary school students in building wireless communication projects using laser diodes (two sessions, class size of ~10 each)
- **Session Co-chair, Compound Semiconductor Week (CSW), Ann Arbor, Michigan (June 2022)**
  Co-chaired five hybrid technical sessions in the CSW 2022
- **Student Volunteer, Compound Semiconductor Week (CSW), Ann Arbor, Michigan (June 2022)**
  Worked as a student volunteer in the CSW 2022 abstract review subcommittee to help with the preparation and proofreading of the official conference booklet and the speaker/poster tracking chart
- **Lab supply purchase coordinator, Zetian Mi Research Group (May 2020-February 2024)**
  In charge of purchase order placements for supplies such as MBE system parts, general lab supplies, and chemical supplies of Zetian Mi Research Group
- **Mentor, Gen II molecular beam epitaxial (MBE) growth (January 2020-July 2024)**
  Provided hands-on training on Gen II molecular beam epitaxy (MBE) operation to several of my research colleagues

- **Organizing Committee Member, IEEE Day Inter-University Project Competition, United International University, Dhaka, Bangladesh (October 2018)**
  Worked as an organizer and judge at the inter-university science project competition in the event of IEEE Day 2018, organized by the IEEE student branch, United International University

TECHNICAL SKILLS:

- **Epitaxial Growth:** Hands-on experience on Gen II molecular beam epitaxial (MBE) growth (~1000 growths) | MBE system maintenance
- **Fabrication techniques/tools:** Cleanroom experience in rapid thermal annealing (RTA), electron beam (e-beam) evaporation, atomic layer deposition (ALD), CMOS silicon oxidation furnace, pre-furnace cleaning (PFC), wet etching, substrate dicing, magnetron sputter deposition, photoresist/liquid dopant spin coating, high-temperature annealing furnace, CMOS phosphorus diffusion furnace
- **Characterization, measurement, and other relevant techniques:**
  Material: Scanning electron microscopy (SEM), energy dispersive spectroscopy (EDS), X-ray diffraction (XRD), X-ray photoelectron spectroscopy (XPS), micro-Raman spectroscopy
  Optical/Electrical: Temperature & power-dependent photoluminescence (PL) measurement, non-contact spectroscopic reflectometry, I-V measurement, electroluminescence (EL) measurement, UV-vis spectrometry
  Chemical: Gas chromatography (GC), photoelectrochemical (PEC) water splitting, linear sweep voltammetry (LSV) scan, chronoamperometry, photocatalytic (PC) water splitting, co-catalyst photodeposition
- **Computer Skills:** JMP, CADENCE, Origin, MATLAB, Large-scale Atomic/Molecular Massively Parallel Simulator (LAMMPS), Quantum Espresso, Silvaco TCAD, Comsol Multiphysics, C/C++

RELEVANT COURSEWORK:

Graduate (M.Sc. and Ph.D.) Courses:
Quantum Phenomena in Nanostructures, Laser Theory, Compound Semiconductor Devices, Semiconductor Characterization Technology, Electronic and Optical Properties of Semiconductors, Solid State Physics, Semiconductor Lasers and LEDs, Properties of Transistors

Undergraduate (B.Sc.) Courses:
Solid State Devices, Compound Semiconductor and Hetero-Junction Devices, Optoelectronics, Analog Integrated Circuit, Electrical Properties of Materials

MEMBERSHIPS:

- The International Society for Optics and Photonics, SPIE (2022-Present)
- OPTICA (2022-Present)
- American Physical Society, APS (2022-Present)
- The Optics Society at the University of Michigan, OSUM (2022-2024)